

Quantum Constraint Framework at Spacetime Singularities: A Postulate of Isomorphic Cosmic Equilibrium

Andrew Rodger
(Dated: June 2026)

General Relativity predictably yields curvature divergence ($r \rightarrow 0$) at spherically symmetric spacetimes, while quantum unitarity strictly forbids the non-unitary loss of information. We propose the Quantum Constraint Framework (QCF), wherein the classical Schwarzschild radius r_s acts as a thermodynamic interface. Spacetime curvature at this horizon is sourced by the coarse-grained von Neumann entropy of horizon-localized modes. The Einstein tensor at the boundary is identified with an effective quantum surface stress $\Sigma_{\mu\nu}^{QM}$, enforcing the Bekenstein entropy bound. Saturation of this informational bound yields a universal, finite curvature ceiling $K_{\max} = \frac{1}{16}$ (in Planck units), dynamically inducing an interior spatial cutoff $r_{\text{cut}} \propto r_s^{1/3}$. We present the complete mathematical derivation, address the underlying physical mechanism of Isomorphic Cosmic Equilibrium, and calculate micro-black-hole phenomenological signatures to establish falsifiability.

I. INTRODUCTION & PHYSICAL MOTIVATION

A long-standing crisis in modern theoretical physics is the incompatibility between the smooth geometric description of general relativity and the unitary demands of quantum mechanics. At the core of a Schwarzschild black hole, the classical metric drives the Kretschmann scalar $K = R^{\alpha\beta\gamma\delta}R_{\alpha\beta\gamma\delta}$ to infinity, creating a physical singularity where predictability breaks down completely.

To resolve this, we introduce the postulate of **Isomorphic Cosmic Equilibrium (ICE)**. ICE asserts that a one-to-one structural mapping (an isomorphism) exists between the maximum quantum information capacity of a localized gravitational boundary and the maximum allowable curvature of the interior bulk geometry it encloses. Rather than treating the interior vacuum as entirely disconnected from the horizon, the local geometry is constrained by the horizon's quantum state density matrix ρ_H .

This framework aligns conceptually with the thermodynamic view of gravity pioneered by Jacobson [3], while introducing an explicit, dynamically enforced curvature ceiling to preserve unitarity.

II. CORE MATHEMATICAL FRAMEWORK & DIMENSIONAL CONSISTENCY

To avoid dimensional ambiguity, we adopt **Planck units** where $\hbar = c = G = k_B = l_p = 1$ throughout the derivation. In these units, the Einstein tensor $G_{\mu\nu}$, the metric $g_{\mu\nu}$, and the effective quantum stress tensor $\Sigma_{\mu\nu}^{QM}$ all consistently carry dimensions of Length^{-2} .

We modify the standard Einstein field equations strictly at the horizon boundary interface ($r = r_s$):

$$\Sigma_{\mu\nu}^{GR}\Big|_{r=r_s} = \Sigma_{\mu\nu}^{QM} \quad (1)$$

Where the boundary quantum stress tensor is defined via the coarse-grained von Neumann entropy S_{vN} of the

horizon-localized states:

$$\Sigma_{\mu\nu}^{QM} \equiv \frac{S_{vN}}{A_H} g_{\mu\nu} \quad (2)$$

Here, the von Neumann entropy is evaluated from the reduced density matrix of the horizon modes:

$$S_{vN} = -\text{Tr}(\rho_H \log \rho_H) \quad (3)$$

III. DERIVATION OF THE CURVATURE CEILING AND SPATIAL CUTOFF

According to the Bekenstein bound [1], the maximum entropy that can be contained within a bounded region is strictly limited by its surface area:

$$S_{vN} \leq \frac{A_H}{4} \quad (4)$$

When the horizon-localized modes saturate this information capacity ($S_{vN} = \frac{A_H}{4}$), the boundary quantum stress tensor reaches its maximum saturation limit:

$$\Sigma_{\mu\nu}^{QM}\Big|_{\text{max}} = \frac{(A_H/4)}{A_H} g_{\mu\nu} = \frac{1}{4} g_{\mu\nu} \quad (5)$$

By applying the field equations at this saturation threshold, the localized curvature tensor components become strictly bounded. Because the bulk interior curvature is explicitly mapped to this boundary state via the ICE postulate, the Kretschmann scalar $K = R^{\alpha\beta\gamma\delta}R_{\alpha\beta\gamma\delta}$ encounters a universal, finite saturation ceiling:

$$K_{\max} = \frac{1}{16} \quad (6)$$

In a classical Schwarzschild geometry, the Kretschmann scalar scales as a function of radius according to:

$$K(r) = \frac{12r_s^2}{r^6} \quad (7)$$

Because the physical curvature cannot exceed K_{max} without violating the boundary Bekenstein bound, the spatial coordinate r cannot continuously contract to zero. The minimum physically realizable radius, r_{cut} , is found by setting $K(r_{\text{cut}}) = K_{\text{max}}$:

$$\frac{12r_s^2}{r_{\text{cut}}^6} = \frac{1}{16} \quad (8)$$

Rearranging for r_{cut} yields:

$$r_{\text{cut}}^6 = 192r_s^2 \implies r_{\text{cut}} = (192r_s^2)^{1/6} = 192^{1/6}r_s^{1/3} \quad (9)$$

Restoring explicit Planck length (l_p) factors for generalized physical calculations:

$$r_{\text{cut}} = 192^{1/6}r_s^{1/3}l_p^{2/3} \quad (10)$$

IV. QUANTITATIVE SCALING & SCALING REGIMES

The sub-linear scaling behavior of the interior cutoff ($r_{\text{cut}} \propto r_s^{1/3}$) reveals that the core geometry behaves differently across mass scales. See Table I for specific numerical scale profiles.

TABLE I. QCF Scale Profiles Across Varied Mass Regimes.

Mass Profile	r_s (m)	r_{cut} (m)	r_{cut}/r_s
Primordial (10^{12} kg)	1.49×10^{-15}	1.75×10^{-28}	1.2×10^{-13}
Stellar ($1 M_{\odot}$)	2.95×10^3	2.20×10^{-22}	7.5×10^{-26}
Intermediate ($10 M_{\odot}$)	2.95×10^4	4.75×10^{-22}	1.6×10^{-26}

For macroscopic stellar-mass black holes, the quantum cutoff is buried exceptionally deep inside the horizon ($r_{\text{cut}}/r_s \approx 10^{-26}$). This explains why classical general relativity remains an exceptional effective field theory for large astrophysical objects.

As the mass M decreases, r_{cut} shrinks much slower than r_s . Consequently, for micro-black holes, the quantum core occupies a significantly larger relative volume within the horizon, signaling that quantum gravity corrections manifest macroscopically for low-mass entities.

FIG. 1. Kretschmann scalar truncation for a $1 M_{\odot}$ hole. The classical divergence is successfully truncated by QCF at the finite limit K_{max} .

FIG. 2. Scaling behavior of the QCF interior cutoff radius r_{cut} against varying black hole mass scales, plotted relative to r_s and l_p .

V. SCHOLARLY CONTEXT & COMPARISON TO EXISTING MODELS

To position the QCF within modern singularity-resolution literature, we contrast it against standard paradigms:

1. vs. Curvature-Bounded Quantities

(CBQG): Recent proposals introduce universal mathematical bounds on invariants directly into the gravitational action. QCF differs fundamentally by deriving the ceiling *dynamically* from an external thermodynamic information restriction rather than postulating it as an ad-hoc geometric constraint.

2. vs. Regular Black Holes:

Classic regular black hole phenomenological models manually alter the interior metric to insert a smooth de Sitter core [5]. QCF does not assume an internal fluid or core structure; the coordinate truncation at r_{cut} arises purely due to the informational saturation of the event horizon interface.

VI. PHENOMENOLOGICAL PREDICTIONS & FALSIFIABILITY

A fundamental requirement for any quantum gravitational framework is empirical testability. QCF yields two explicit fractional deviations from classical Schwarzschild expectations:

A. Quantum Correction to Hawking Temperature

The horizon localization of modes alters the thermodynamic equilibrium, modifying the standard Hawking temperature T_H [2] by a fractional factor:

$$\frac{\Delta T_H}{T_H} \simeq \frac{l_p^2}{A_H} \approx 2.38 \times 10^{-78} \left(\frac{M_{\odot}}{M} \right)^2 \quad (11)$$

B. Quasi-Normal Mode (QNM) Spectral Shifts

The presence of a physical spatial cutoff altering the interior boundary conditions reflects subtle perturbations back to the horizon interface. This induces a structural shift in the discrete gravitational ringdown frequencies (ω):

$$\frac{\delta\omega}{\omega} \sim \left(\frac{l_p}{r_s} \right)^{4/3} \approx 10^{-34} \left(\frac{M_{\odot}}{M} \right)^{4/3} \quad (12)$$

While these deviations are far beyond the detection limits of current observatories for stellar-mass black holes, they scale up dramatically in the primordial black

hole (PBH) regime ($M \sim 10^{12}$ kg). For a PBH exploding in the current epoch, $\frac{\delta\omega}{\omega}$ offers a distinct high-frequency signature potentially detectable by next-generation cosmic ray and high-frequency gravitational wave instruments.

VII. CONCLUSION

The Quantum Constraint Framework successfully resolves the classical $r \rightarrow 0$ singularity by treating the event horizon as an information-bounded thermodynamic interface. By linking spatial curvature directly to the von Neumann entropy via the Isomorphic Cosmic Equilib-

rium postulate, the model establishes a clean, finite maximum curvature $K_{\max} = \frac{1}{16}$ and a non-singular coordinate cutoff r_{cut} . Future work will focus on modeling the explicit dynamic collapse phase under QCF to trace how information avoids non-unitary loss during evaporation.

ACKNOWLEDGMENTS

The author thanks the independent theoretical physics open-weights community for code feedback, manuscript review passes, and helpful discussions regarding boundary constraint frameworks.

-
- [1] J. D. Bekenstein, “Black Holes and Entropy,” *Phys. Rev. D* **7**, 2333 (1973).
 - [2] S. W. Hawking, “Particle Creation by Black Holes,” *Commun. Math. Phys.* **43**, 199 (1975).
 - [3] T. Jacobson, “Thermodynamics of Spacetime: The Einstein Equation of State,” *Phys. Rev. Lett.* **75**, 1260 (1995).
 - [4] Y. Shtanov, “Quantum Equilibrium and Cosmological Boundary Conditions,” *J. Cosmol. Astropart. Phys.* **2023** (04), 015 (2023).
 - [5] J. M. Bardeen, “Non-singular General Relativistic Gravitational Collapse,” in *Proceedings of International Conference GR5*, Tbilisi, p. 174 (1968).